

WANG HOUTIAN

Undergraduate Physics Student · Nanyang Technological University

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EDUCATION

Nanyang Technological University, Singapore

Expected 2028

Bachelor of Science in Physics

CGPA: 4.38/5.00

Transferred from Materials Science and Engineering to Physics in 2026.

RESEARCH EXPERIENCE

From the Quantum Particle to Nonrelativistic Quantum Fields: A Mathematical Tour

August 2026-Present

URECA Project · Supervisor: Prof. François Gay-Balmaz · NTU

- Developing a mathematically structured route from single-particle quantum mechanics to nonrelativistic quantum field theory.
- Constructing an explicit dictionary between one-particle and second-quantized formulations, including bosonic and fermionic Fock spaces, creation and annihilation operators, quantum fields, and possible extensions to symmetry.

Memory Effects in Stochastic Thermodynamics

May 2026-Present

Undergraduate Research Project · Supervisor: Prof. François Gay-Balmaz · NTU

- Investigating memory effects in stochastic thermodynamics through a variational formulation proposed by Prof. François Gay-Balmaz.
- Applying the framework to the full non-Markovian system, examining fluctuation-dissipation consistency, stationary behavior, and entropy production.

RESEARCH INTERESTS

Undergraduate focus: Non-Markovian stochastic dynamics; stochastic thermodynamics; nonequilibrium statistical mechanics.

Future directions: Mathematical physics, symmetry, and representations; quantum field theory, holography, and AdS/CFT; quantum information theory.

SELECTED ACADEMIC WRITING

Analytical Mechanics

Available on website

Structured notes on Lagrangian and Hamiltonian mechanics.

Classical Field Theory

In preparation

Quantum Mechanics

In preparation

Mathematical Methods for Physics

In preparation